A Unified SME ↔ Exotic-Potential Framework for CPT Symmetry Tests

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Research Question

Exotic bosons predicted across many BSM scenarios (axion-like particles, dark-sector mediators, Lorentz/CPT-violating fields) generate spin-dependent forces between fermions. Over 100 experiments since 2024 have placed bounds on their couplings, but matter-sector and antimatter-sector results have never been systematically compared on common footing — despite this being a direct CPT test.

Framework

The Standard Model Extension (SME) parameterises Lorentz/CPT violation via fixed background fields (e.g. b_μ, $H_{\mu\nu}$) in the fermion Lagrangian. A Foldy–Wouthuysen transformation maps these SME coefficients onto the non-relativistic Dobrescu–Mocioiu potential basis (V1–V16), which is the language most exotic-force experiments report bounds in. SPINDEP then compiles published constraint datasets, matches matter datasets to their antimatter conjugates (e.g. e-e vs e-e+), interpolates both onto a shared interaction-range λ grid, and computes an asymmetry parameter

$$A_\alpha(\lambda) = [g\_matter(\lambda) – g\_antimatter(\lambda)] / [g\_matter(\lambda) + g\_antimatter(\lambda)]$$

with a χ^2 test quantifying whether any observed asymmetry is statistically significant.

Status & Current Results

- SME → potential-basis derivations (Foldy–Wouthuysen) verified for b_μ and $H_{\mu\nu}$ coefficients; $d_{\mu\nu}$ re-verification in progress.
- 273 experimental constraint datasets compiled and classified across gAgA, gsgs, gVgV, gpgp and gpgs couplings.
- 10 matter–antimatter pairs successfully matched so far, spanning lepton-lepton and lepton-nucleon sectors.
- Mean $|A_\alpha|$ across matched pairs ranges from 0.33 to 1.00 — i.e. matter and antimatter bounds on the same coupling currently differ substantially in strength, purely as a reflection of which sector has been probed more precisely.
- Naive χ^2 test (uniform 10% fractional uncertainty, 300 interpolated grid points as dof) reports $p \approx 0$ for every pair. Correcting for this via an autocorrelation-length estimate of the effective dof (6–21 per pair, not 300) still gives $p \approx 0$ for every pair — the significance survives the most obvious objection, which sharpens rather than resolves the open question below.

What I'd Value Feedback On

- Experimental grounding: do the matter/antimatter dataset pairings correctly reflect the underlying experiments, and are there important CPT-test results I'm missing?
- Theoretical framing: is A_α over the SME-derived potential basis a well-motivated observable, and how should these bounds connect to concrete BSM model space?
- Statistical methodology: after correcting for correlated grid points via an effective dof (autocorrelation length of the residuals, typically 6–21 vs. the naive 300), every pair still shows $p \approx 0$. Is this correction the right one, and if it's applied correctly, what else could be driving significance this extreme — the per-point uncertainty model, or something about comparing one-sided bounds at all?

Full derivations, dataset registry, and code: github.com/oyewodayo/spin-dependent-exotic-interactions-SME